

Essentials of Data Science

Lecture-1
CS5113

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Lecture Plan

- Data
- Data types
- What is Data Science?
- Homework

Data Data Everywhere

In today's world, especially here in India, **we're surrounded by data**. Every day, we **generate, transmit**, and often **give away** data without realizing it. Each click, scroll, tap, text, or voice note can become a variable influencing business decisions and economies.

[Current World Population: 8,298,979,488](#)

OCT
2025

POPULATION ESSENTIALS

DEMOGRAPHICS AND OTHER KEY INDICATORS



TOTAL
POPULATION



8.25
BILLION



FEMALE
POPULATION



49.7%

we
are
social

MALE
POPULATION



50.3%



YEAR-ON-YEAR CHANGE
IN TOTAL POPULATION



+0.8%
+69 MILLION



MEDIAN AGE OF
THE POPULATION



30.9

URBAN
POPULATION



58.4%



POPULATION DENSITY
(PEOPLE PER KM²)



63.1



OVERALL LITERACY
(ADULTS AGED 15+)



87.4%

we
are
social

FEMALE LITERACY
(ADULTS AGED 15+)



84.1%

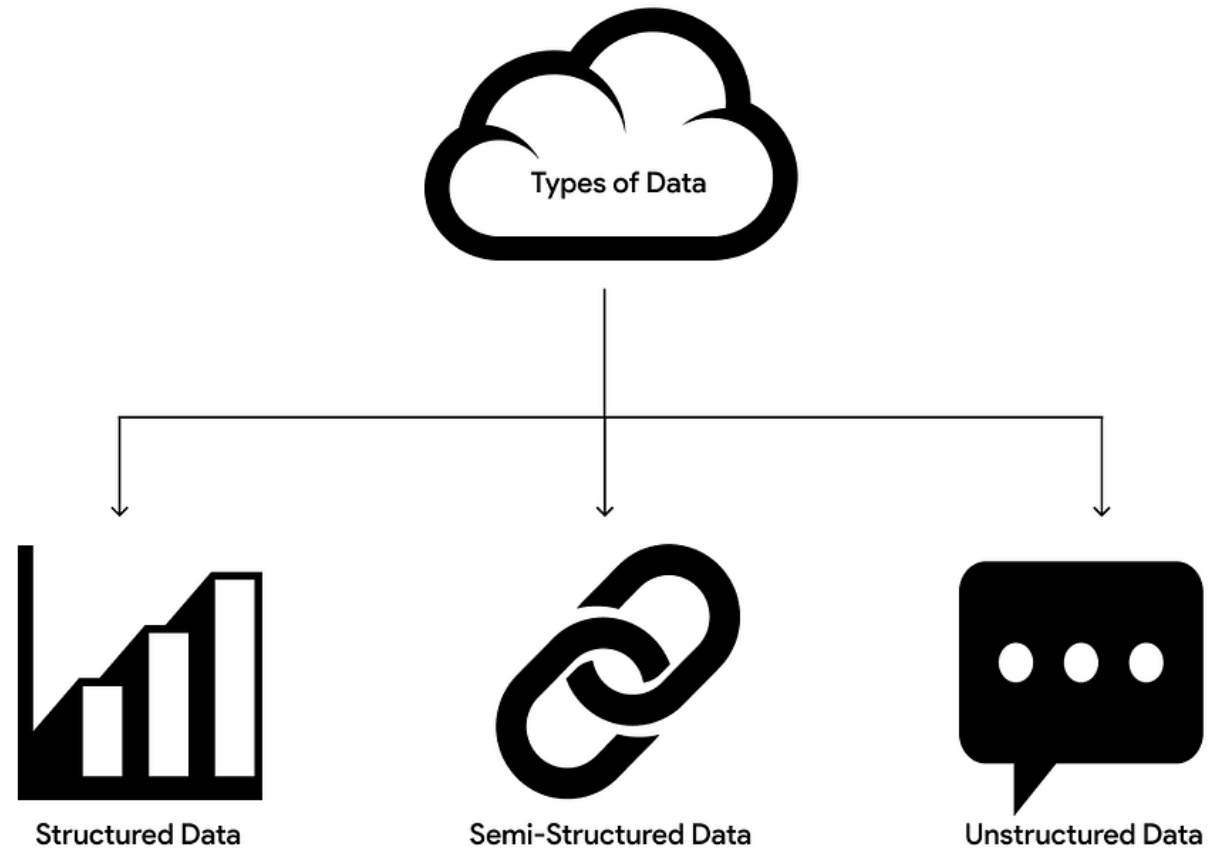


MALE LITERACY
(ADULTS AGED 15+)

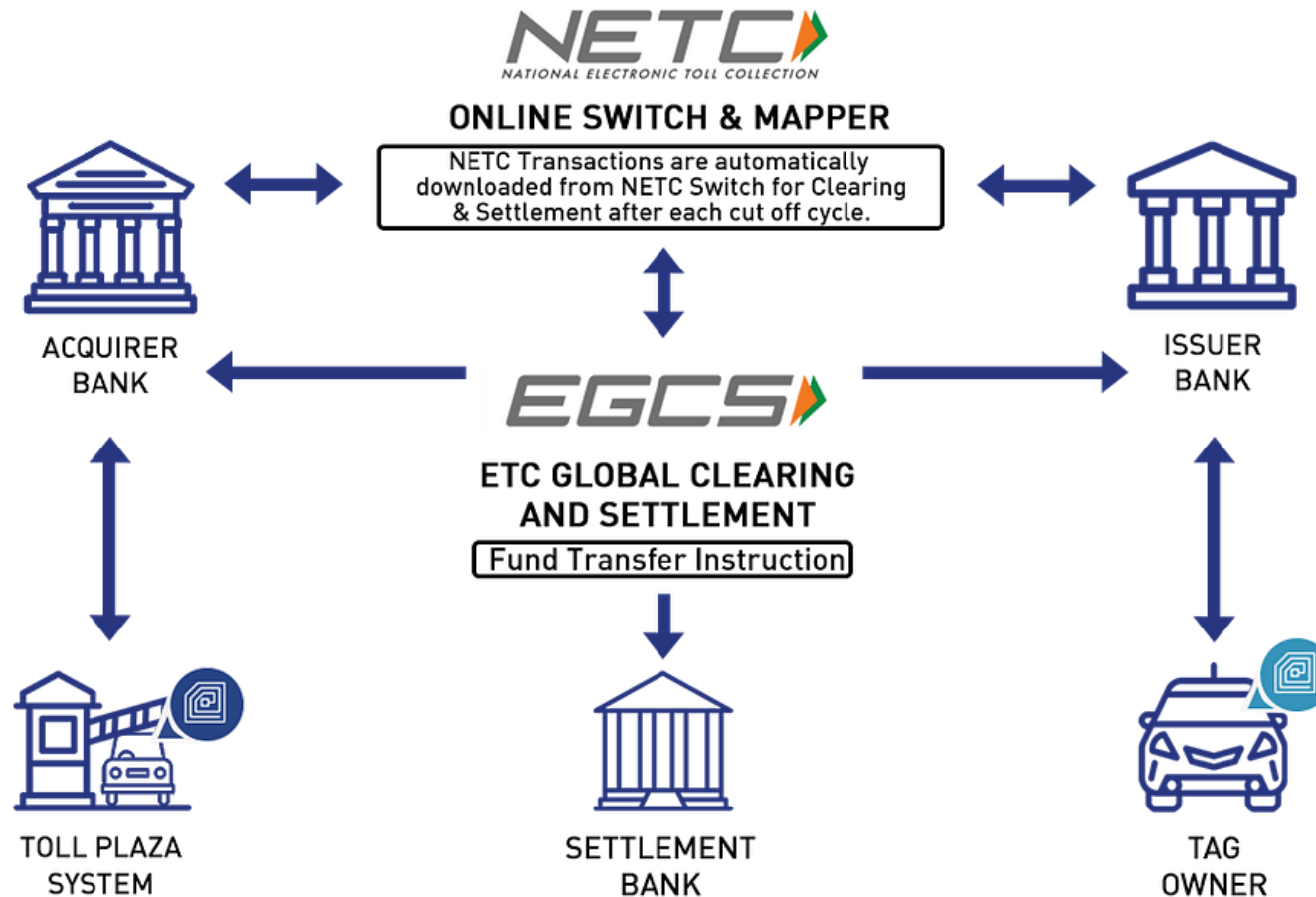


90.6%

Types of Data



Data generation every minute



Data generation every minute

Semi-structured Data



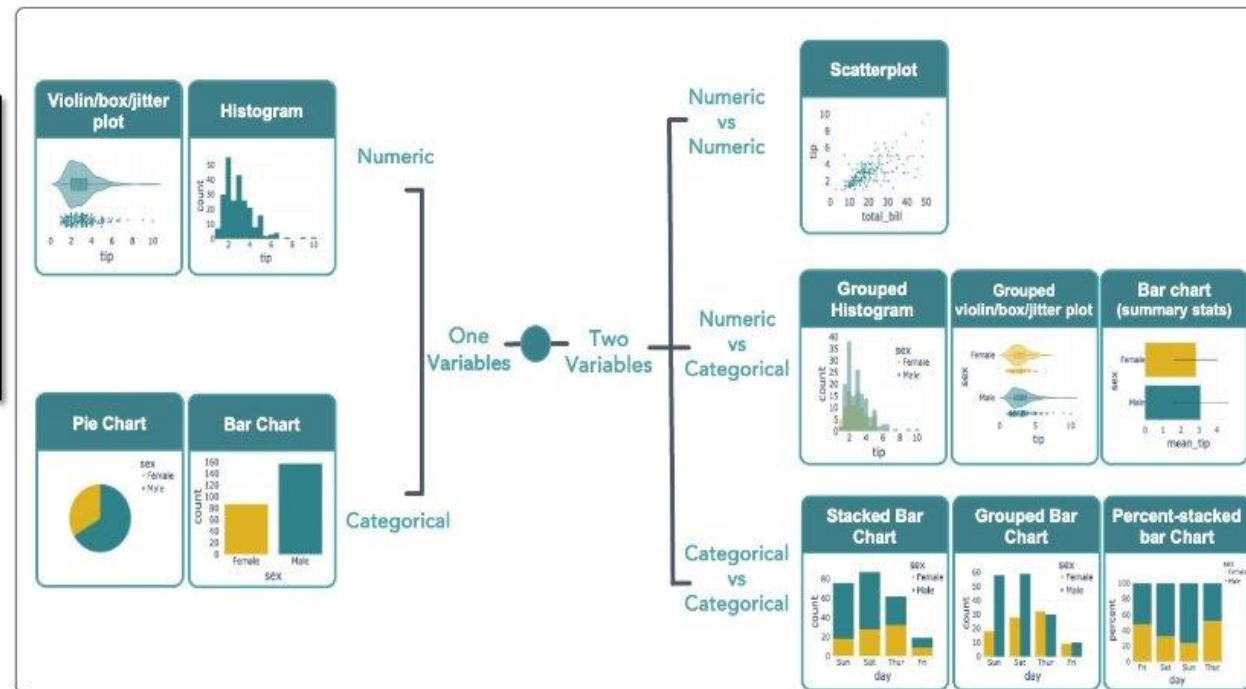
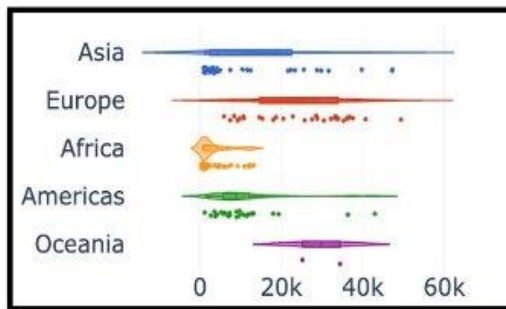
How does data look like?

Practice: Identify visualization

1. How does country GDP per capita vary across continents?

country	continent	lifeExp	gdpPercap	income_group
Afghanistan	Asia	43.828	974.58033	Low & Middle Income
Albania	Europe	76.423	5937.0295	Low & Middle Income
Argentina	Americas	75.32	12779.379	Low & Middle Income
Austria	Europe	79.829	36126.4927	High Income
Bahrain	Asia	75.635	29796.04	High Income

1.



Type of data vs Data types

Type	On Disk	In Python
tabular	CSV	DataFrame
hierarchical	JSON	dict
textual	plaintext	str
geospatial	???	???

Questions to ask?

- What am I hoping to understand?
- What do I need to know to make a certain business decision?
- What story is this data telling?
- What do the relationships between variables mean for ____ at my company?
- What if ____ changed? Which variables, trends, or forecasts would be impacted?
- What needs to change in the data to get the desired outcome?
- Why does the data trend in this direction, and what does it mean for the future?
- How can I further analyze the data to get the answers needed to make important decisions

Data Driven decision making framework

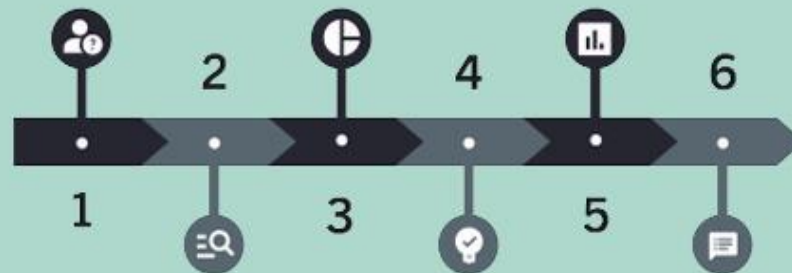


BUSINESS INSIGHT

The Data-Driven Decision-Making Framework

There are various tools and frameworks that can be beneficial when using data. One is the data-driven decision-making framework, which is explained in the Harvard Online course Data Science for Business. This framework presents six steps for using data to drive business decisions:

Data-Driven Decision-Making Framework



- 1. Understand the business problem:** What are you looking to understand or accomplish?
- 2. Wrangle data:** Clean, validate, and organize the data.
- 3. Create visualizations:** Present the data in a way that shows trends and relationships of interest.
- 4. Generate hypotheses:** Formulate predictions based on emerging trends.
- 5. Conduct analysis:** Run statistical tests to determine if your hypotheses are correct.
- 6. Communicate results:** Present your findings in the context of the original business problem.

Understanding data frameworks like this one gives you the capability to take a raw dataset, interpret its story, and use it to answer relevant business questions.

Practical Homework

1. Import a data file

[datasets/titanic.csv at master · datasciencedojo/datasets · GitHub](#)

2. Ask questions
3. Summarizing and visualizing tabular data

Submission link: [Homework-1 \(Essentials of Data Science\)-Fall Semester 2026 – Fill out form](#)

Deadline: 10 August 2026, 11:59 PM

References

[Social Media Usage Statistics \(2026\): Key Facts & Insights](#)

[datasets/titanic.csv at master · datasciencedojo/datasets · GitHub](#)